Article title: Heterogeneous chromatin and transcription-factor signatures across functionally supported Gli3 limb enhancers in embryonic mouse limb

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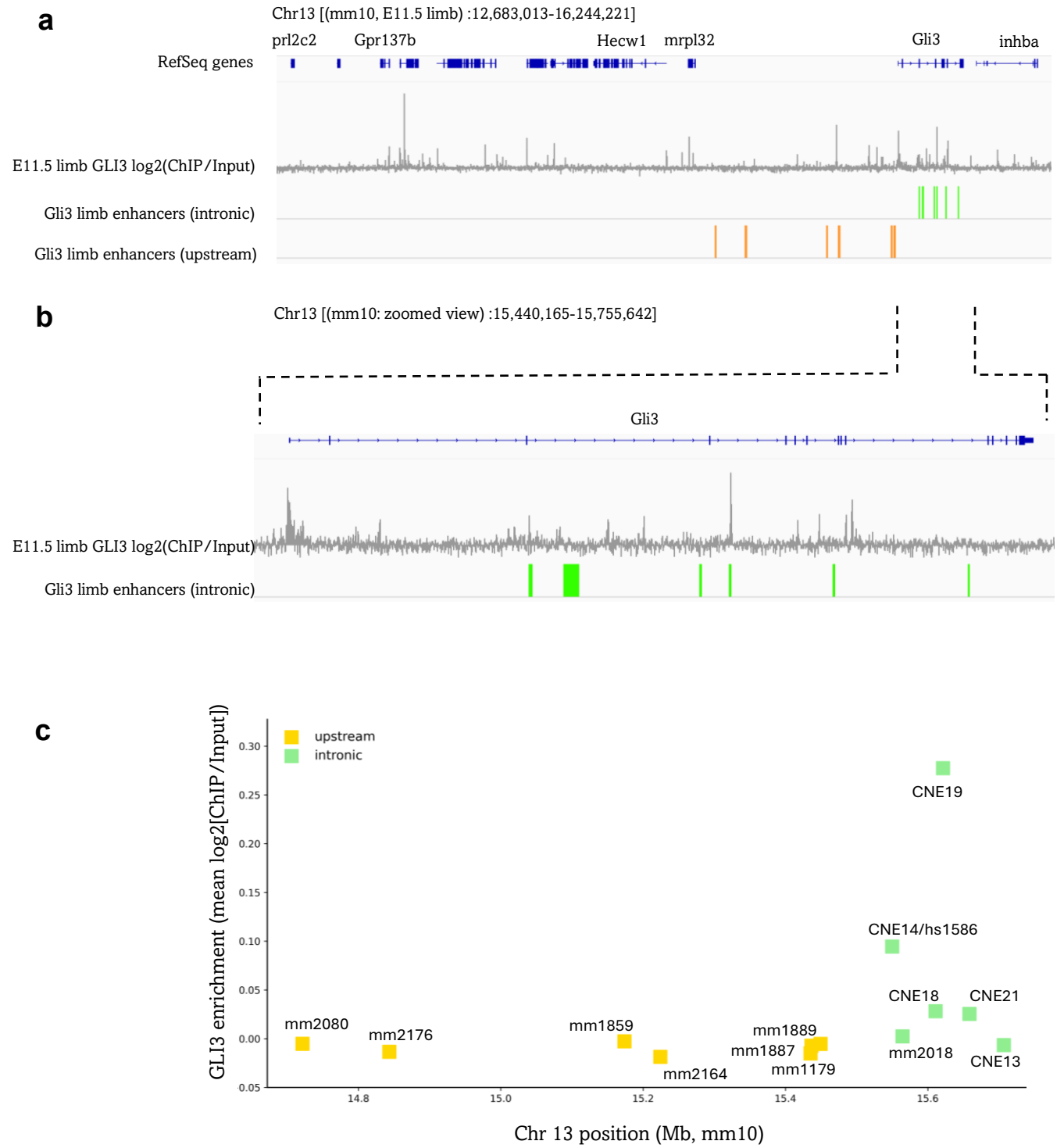
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Online Resource 1: Supplementary Figures S1 and S2

Supplementary Figure S1



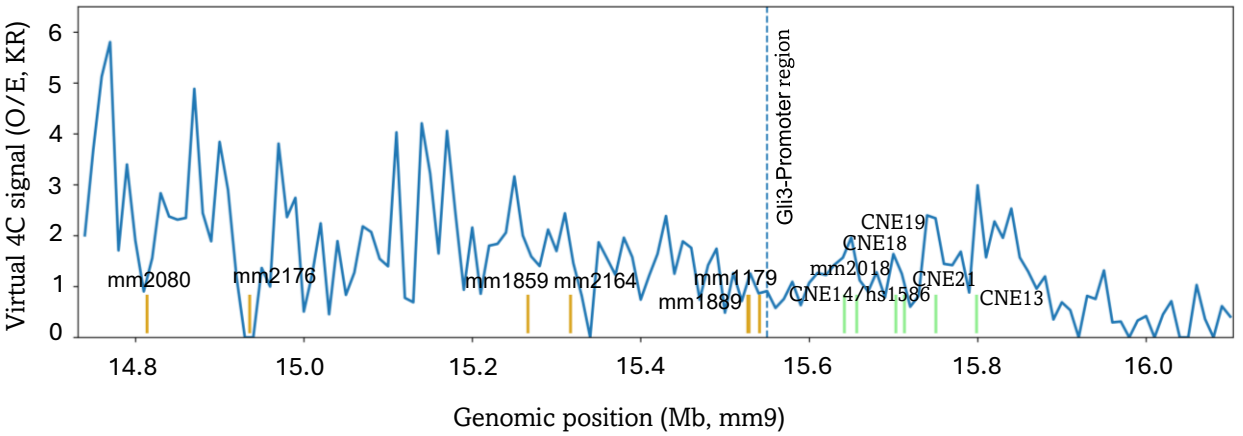
Supplementary Figure S1. GLI3 ChIP-seq signal across the curated Gli3 limb-enhancer set (mm10, WT E11.5 limb).

(a) Genome-browser view of the broader *Gli3* locus (mm10) showing GLI3 ChIP-seq enrichment in WT E11.5 limb as a log2(ChIP/Input) bigWig track. Reads from three GLI3 ChIP-seq runs (SRR11907926, SRR11907927, SRR11907928) were aligned to mm10 using Bowtie2, merged, and normalized to CPM prior to log2(ChIP/Input) calculation. Track display ranges were set with minimum = -0.15 and maximum = 1.0. The curated *Gli3* limb enhancer set is displayed as color-coded intervals (upstream in gold; intronic in green). Gene models are shown for positional reference. Display range was fixed across views to allow consistent visual comparison.

(b) Zoomed-in genome-browser view highlighting representative enhancer intervals at higher local resolution using the same GLI3 log2(ChIP/Input) track and scale conventions as in panel a. This view enables direct visual assessment of GLI3 ChIP-seq signal across individual enhancer elements and shows higher signal at a subset of intronic enhancers, whereas several other elements exhibit comparatively weak or near-background signal.

(c) Quantification of GLI3 ChIP-seq signal across individual Gli3 enhancers (mm10). For each BED-defined enhancer interval, mean GLI3 enrichment was calculated as the average log2(ChIP/Input) signal across the full enhancer length. Enhancers are plotted by genomic position (x-axis) against mean GLI3 enrichment (y-axis) and color-coded as upstream or intronic. GLI3 signal varies substantially among the 13 enhancers, with the strongest mean enrichment observed at CNE19 (~0.28), followed by CNE14/hs1586 (~0.095), CNE18 (~0.028), and CNE21 (~0.025), whereas several other enhancers show weak or near-background mean signal. The numerical values underlying panel c are provided in Table S3 (Online Resource 2).

Supplementary Figure S2



Supplementary Figure S2. Virtual 4C from the *Gli3* promoter in E11.5 hindlimb Hi-C (mm9).

Virtual 4C was computed from KR-balanced Hi-C matrices from mouse E11.5 hindlimb buds (GEO GSE116791; NCBI37/mm9). The viewpoint was the 10-kb bin containing the *Gli3* promoter/TSS (chr13:15,550,000–15,560,000; dashed line). The profile shows observed/expected (O/E), KR-normalized contact enrichment between the promoter-containing bin and successive 10-kb bins across chr13:14,732,034–16,103,316 (mm9). Vertical ticks mark the 13 curated *Gli3* limb enhancers analyzed in this study (light green, intronic; gold, upstream). The profile provides a promoter-centered view of contact enrichment across the *Gli3* region; these population-averaged Hi-C contacts are descriptive and do not by themselves establish direct enhancer–promoter looping or functional interaction. Underlying signal values and enhancer coordinates are provided in Tables S4a and S4b (Online Resource 2). TSS, transcription start site; O/E, observed/expected; KR, Knight–Ruiz matrix balancing; GEO, Gene Expression Omnibus.